

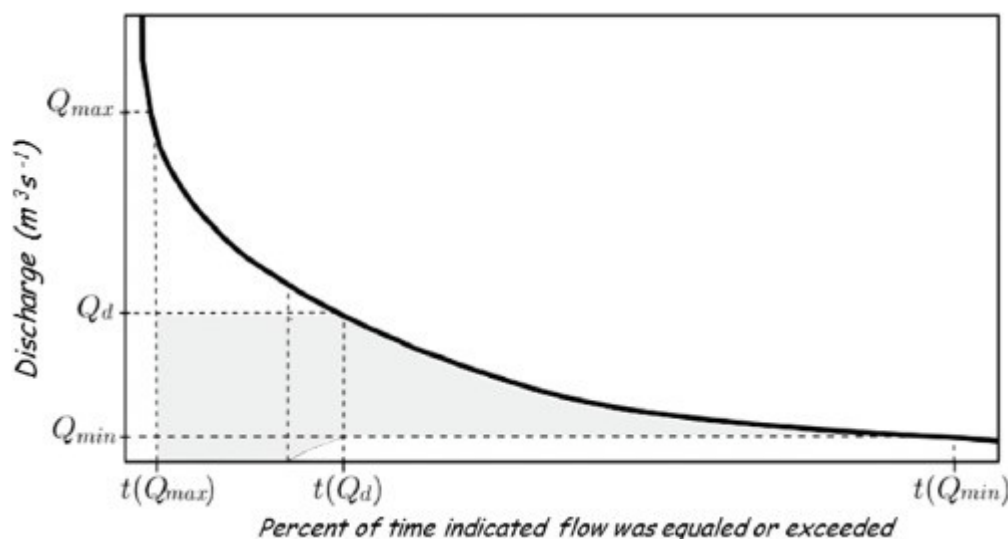
1. What do you mean by Flow Duration Curve? How do you plot it?

Answer:

A Flow Duration Curve (FDC) is a plot showing the percentage of time that a given discharge is equaled or exceeded. It is widely used to determine the design discharge and hence the installed capacity of a Run-of-River hydropower plant. It is constructed by following procedure:

Procedure:

1. Collect long-term streamflow data.
2. Arrange discharges in descending order.
3. Compute exceedance probability.
4. Plot the Flow Duration Curve.
5. Select the design discharge (e.g., Q_{40} , Q_{50} , or Q_{60}).
6. Calculate installed capacity using: $P = \rho g Q H \eta$.



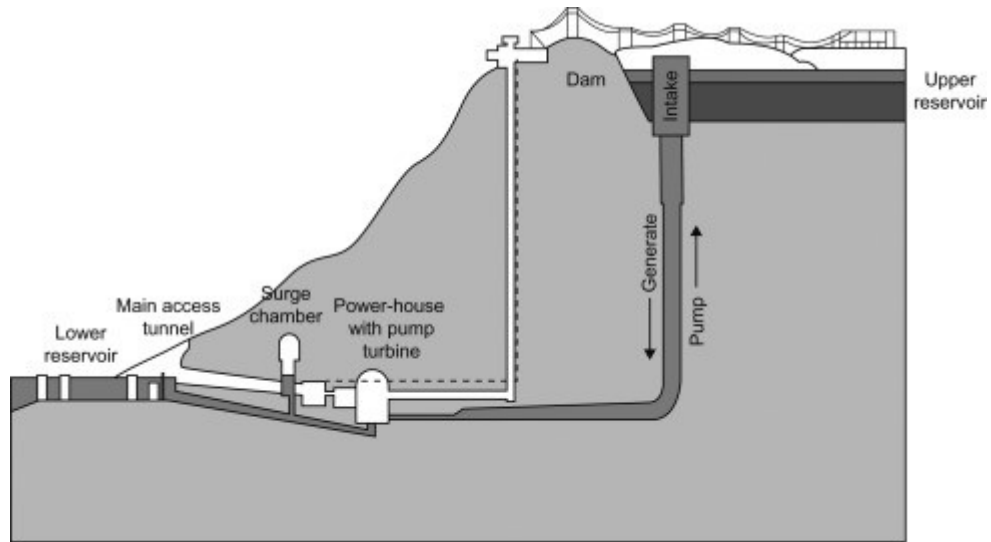
The three important application of flow duration curve is stated below:

- Hydropower planning and design:**
Flow Duration Curve is used to estimate the amount of water available for power generation and selection of the installed capacity of hydropower plants.
- Water resources planning and design:**
Flow Duration Curve also helps to assess the reliability of streamflow for irrigation, municipal water supply and estimate the storage capacity of reservoirs.
- Assessment of stream flow characteristics:**
Flow duration curve is also useful for analyzing the flows of river for river management, environmental flow studies and drought analysis.

2. Discuss in brief about basic design, hydraulic design and structural design of hydroelectric project components or structures.

Answer:

The design of a hydroelectric project involves basic design, hydraulic design, and structural design of various components to ensure safe, efficient, and economical generation of power.



i. Basic Design

Basic design is the initial planning and sizing of the major components of a hydropower project. It is based on available water, topography, geology, power demand, and project economics.

It includes:

- Assessment of catchment area, river discharge, dependable flow and design flood.
- Determination of design head and installed capacity.
- Selection of project layout and type of scheme—run-of-river, storage, or peaking.
- Preliminary sizing and alignment of dam/weir, intake, waterways, surge tank, powerhouse and tailrace.
- Selection of suitable turbines, generators and other electromechanical equipment.
- Consideration of access roads, sediment control, construction requirements and environmental aspects.

ii. Hydraulic Design

Hydraulic design deals with the movement and control of water through the hydropower system with minimum energy loss and without causing unsafe hydraulic conditions.

It includes:

- Determination of design discharge and flood discharge.
- Design of intake, trash rack, approach channel and settling basin.
- Hydraulic design of headrace tunnel/canal, penstock, surge tank and tailrace.
- Calculation of head losses due to friction, bends, valves, transitions and other structures.
- Checking water velocities, pressure conditions and cavitation.
- Design of spillways, energy-dissipation arrangements and outlet works.
- Analysis of water hammer and transient flow in pressure waterways.

iii. Structural Design

Structural design ensures that all civil structures can safely resist water pressure, earth pressure, seismic forces, self-weight and other loads throughout their service life.

It includes:

- Structural design of dam/weir, intake, spillway and stilling basin.
- Design of tunnels, canals, aqueducts, penstocks and surge tanks.
- Design of powerhouse, foundations, retaining walls and tailrace structures.

➤ Consideration of loads such as:

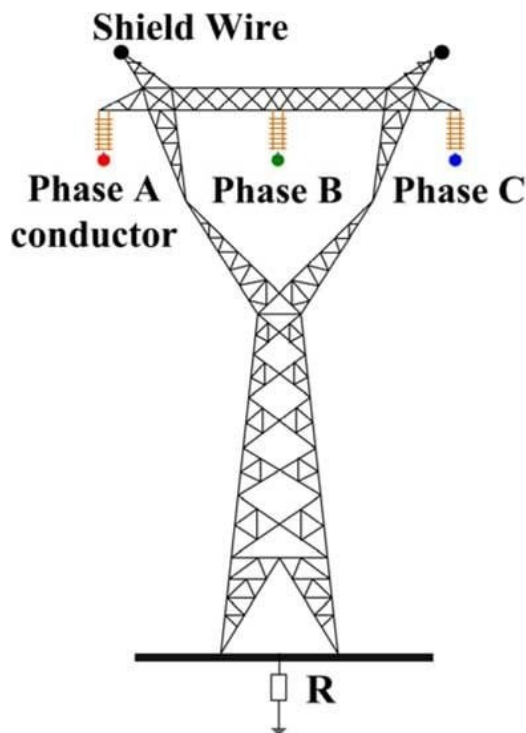
- Dead load
- Hydrostatic and hydrodynamic pressure
- Earth and rock pressure
- Uplift pressure
- Seismic forces
- Temperature and shrinkage effects
- Construction and maintenance loads

➤ Checking structures for stability, strength, deformation, cracking and durability.

3. **What do you understand by single circle and double circle circuit of high voltage transmission line? What is the maximum voltage of transmission line existing in Nepal?**

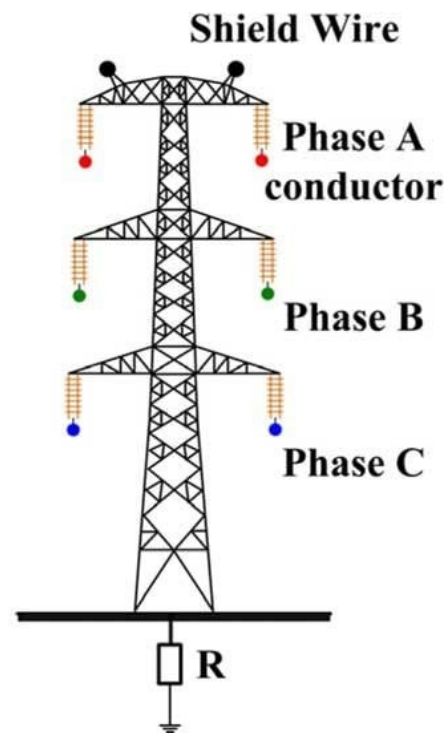
Answer:

A circuit of a transmission line consists of one complete set of conductors required to transmit three-phase electrical power.



(a)

a. Single circuit transmission line



(b)

b. Double circuit transmission line

i. **Single-Circuit Transmission Line**

A single-circuit transmission line has one set of three phase conductors mounted on the transmission towers.

- It carries one three-phase electrical circuit.
- It generally requires fewer conductors and simpler tower arrangements.
- It has lower initial construction cost.
- However, if the circuit fails, the entire power transfer through that line is interrupted.
- It is suitable where the required power-transfer capacity and reliability are relatively moderate.

ii. **Double-Circuit Transmission Line**

A double-circuit transmission line has two independent three-phase circuits mounted on the same tower or tower structure.

- It contains six phase conductors (three for each circuit).
- It can transmit more power than a single circuit.
- It provides greater reliability and continuity of supply because one circuit can continue operating if the other is taken out of service.
- It requires larger and stronger towers and generally has a higher construction cost.
- It is commonly used for important transmission corridors and high-power transfer.

Differences:

Basis	Single Circuit	Double Circuit
Number of circuits	1	2
Phase conductors	3	6
Power-transfer capacity	Lower	Higher
Reliability	Lower	Higher
Tower requirement	Smaller	Larger/stronger
Cost	Relatively low	Relatively high

Maximum transmission voltage in Nepal:

At present, 400 kV is the highest transmission voltage level being used/implemented in Nepal's transmission system. For example, the Dhalkebar–Muzaffarpur 400 kV cross-border transmission line is a 400 kV line, and Nepal Electricity Authority documents also identify 400 kV systems.

Furthermore, the Department of Electricity Development's current records include several 400 kV transmission projects, including the Upper Arun, Dudhkoshi, and other 400 kV corridors.

Therefore: Maximum transmission voltage in Nepal = 400 kV.

4. When was the first hydropower commissioned in Nepal? Describe the hydropower development history in Nepal.

Answer:

- The history of hydroelectricity generation in Nepal began on 22 May 1911 A.D. (9 Jestha 1968 B.S.) with the commissioning of the 500 kW Pharping Hydropower Station, also known as *Chandra Jyoti*.
- It was inaugurated by King Prithvi Bir Bikram Shah during the premiership of Chandra Shumsher Rana
- It is recognized by the Nepal Electricity Authority as Nepal's first power station.



- The early development of hydropower was very slow.
- After Pharping, the 640 kW Sundarijal plant was commissioned in 1936, followed by Panauti in 1965 and major projects such as Trishuli in 1967.
- During this period, electricity supply was very limited and was concentrated mainly in Kathmandu and nearby areas.
- Hydropower development was controlled by the government and was intended primarily for administrative centers, palaces, and limited urban use.
- During the 1960s–1980s, Nepal began developing larger hydropower projects with assistance from international development partners.
- Important projects included Panauti, Trishuli, Sunkoshi, Gandak, Kulekhani and Devighat. Kulekhani-I, commissioned in 1982 with 60 MW capacity, was particularly important because it was a storage-type hydropower project.
- The First Five-Year Plan, covering 1956–1961, gave importance to projects such as Trishuli, Panauti, Thadokhola, Seti, and Tinau.
- The main objective was to provide electricity for industrialization, irrigation, urban development, and public services.
- The Nepal Electricity Authority (NEA) was established in 1985 through the merger of several electricity-related organizations.
- This strengthened the coordinated development, transmission and distribution of electricity.
- After the restoration of democracy and economic liberalization in the 1990s, the private sector began participating significantly in hydropower development.
- Khimti Hydropower Project (60 MW) became an important milestone in private-sector participation and was commissioned in 2000.

The 2000s saw the development of larger projects such as:

- i. Kali Gandaki 'A' – 144 MW (2002)
- ii. Middle Marsyangdi – 70 MW (2008)
- iii. Upper Trishuli 3A – 60 MW
- iv. Upper Tamakoshi – 456 MW (2021)

The 456 MW Upper Tamakoshi Hydroelectric Project marked a major milestone as Nepal's largest hydropower project by installed capacity at the time of commissioning.

In recent years, hydropower development has accelerated considerably, with increasing participation from private developers and foreign investment, expansion of the national transmission grid, and development of large projects for both domestic consumption and electricity export.

5. Describe the state of project studies and elements of field investigation required for these studies.

Answer:

For a hydropower project, studies are carried out in different stages, with the level of investigation increasing as the project progresses. The major objective is to establish the technical feasibility, economic viability, environmental acceptability and safety of the project.

Stages of Project Studies

a. Reconnaissance Study

It is the initial investigation carried out to identify promising project sites.

- Study of available maps, satellite images and existing reports.
- Preliminary assessment of topography, geology and hydrology.
- Assessment of accessibility and major environmental constraints.

b. Pre-feasibility Study

It determines whether the identified project is sufficiently promising for detailed investigation.

- Comparison of alternative project layouts.
- Preliminary hydrological and geological assessment.
- Preliminary power and energy estimation.

- Identification of major technical, environmental and social issues.

c. Feasibility Study

This is a detailed study to establish whether the project can practically and economically be developed.

- Detailed topographical survey.
- Detailed hydrological and sediment studies.
- Geological and geotechnical investigations.
- Environmental and social impact assessment.
- Detailed cost estimate and economic/financial analysis.

d. Detailed Project Report (DPR) / Detailed Design

The selected project alternative is developed to a level suitable for construction.

- Final design and dimensions of all major structures.
- Detailed geological and geotechnical investigations.
- Detailed structural and hydraulic design.
- Detailed quantity and cost estimates.

Elements of Field Investigation:

a. Topographical Investigation

- Preparation of detailed contour maps.
- Longitudinal and cross-sectional surveys of rivers.
- Survey of dam, intake, powerhouse, tunnel and canal sites.
- Survey for access roads, transmission lines and construction facilities.

b. Hydrological Investigation

- Measurement of river discharge and water levels.
- Collection of rainfall and climatic data.
- Flood-frequency analysis.
- Assessment of sediment load and water availability.

c. Geological Investigation

- Geological mapping of the project area.
- Identification of rock types, faults, folds, joints and discontinuities.
- Geological investigation along tunnels and pressure waterways.
- Identification of landslide and unstable zones.

d. Geotechnical Investigation

- Collection of soil and rock samples.
- Determination of bearing capacity and shear strength.
- Permeability and groundwater investigation.
- Rock-mass classification and foundation assessment.

e. Seismic Investigation

- Study of regional seismicity and active faults.
- Assessment of earthquake hazards.
- Determination of appropriate seismic design parameters.

6. What is difference between economic and financial analysis?

Answer:

Economic Analysis	Financial Analysis
Economic analysis evaluates the project from the viewpoint of the economy/society as a whole.	Financial analysis evaluates the project from the viewpoint of the project owner/investor.
It determines whether the project provides overall economic and social benefits.	It determines whether the project is financially profitable and financially sustainable.
Economic analysis represents the national or societal perspective.	Financial analysis represents the individual/project perspective.
It uses economic/shadow prices where appropriate to reflect the real opportunity cost of resources.	It uses actual market/financial prices.
It considers external benefits and costs such as environmental and social impacts where they can be valued.	It usually considers only costs and benefits directly affecting the project.
Financing structure is generally not the main concern.	Financial analysis considers loans, interest, equity, debt repayment and financing costs.
Economic rate of return (ERR), economic NPV, benefit-cost ratio, etc.	Financial rate of return (FIRR), financial NPV, cash flow, debt-service capacity, etc.

7. What is procedure generally to be followed as per existing environmental regulation for conducting initial environmental environment (IEE) of a small hydropower project.

Answer:

According to Environment Protection Act (EPA) 2076 and Environment Protection Rules (EPR) 2077, IEE is defined as "An analytical study or evaluation conducted to determine whether a proposed project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts." It is generally done for medium scale projects that has moderate effect on environment.

The general procedure required to conduct and approve an IEE for such a project involves the following steps:

i. Preparation and Approval of the Terms of Reference (ToR / Work Schedule)

- **Drafting the ToR:** The project proponent must first prepare a detailed Work Schedule (ToR/कार्यमूची) in the format prescribed in Schedule 7 of the Regulations. This document outlines the scope of the study, environmental parameters to be analyzed, methodologies for data collection, expert team composition, budget, and study timeline.
- **Approval by Concerned Body:** The proponent submits this ToR to the Concerned Body (e.g., the Department of Electricity Development / Ministry of Energy, Water Resources and Irrigation at the federal level, or the designated provincial/local department depending on project jurisdiction). The concerned body reviews, modifies if necessary, and officially approves the ToR.

ii. Field Investigations and Baseline Data Collection

Following ToR approval, the proponent's multidisciplinary team conducts field surveys to gather qualitative and quantitative baseline data:

- **Physical Environment:** Studies on hydrology (water flow), water quality, geology, landslide risks, air, soil, and noise levels.
- **Biological Environment:** Assessment of local forests, vegetation loss, wildlife, habitats, and aquatic life—especially fish species and migration patterns.
- **Socio-economic & Cultural Environment:** Data on local populations, land acquisition, livelihoods, displacement, and nearby historical or religious structures.

iii. Public Notice and Local Consultations

To ensure public accountability and local participation:

- **15-day Public Notice:** The proponent must publish a 15-day public notice in a national daily newspaper describing the project's features and potential environmental impacts.
- **Pasting Notices & Muchulka:** Copies of this notice must be posted on the notice boards of the project-affected Rural Municipality/Municipality, local schools, hospitals, and health posts. A formal deed of public inquiry (**Muchulka**) must be signed by local witnesses to legally certify that the notice was pasted.
- **Collection of Suggestions:** The proponent gathers feedback, concerns, and suggestions from local stakeholders during this 15-day period.

iv. **Detailed Analysis of Alternatives and Mitigation Measures**

- **Alternative Analysis:** The proponent must evaluate possible project alternatives (including design, construction technology, location of key structures, and scale) to minimize forest and community impacts.
- **Drafting the EMP:** Proponents prepare a structured Environmental Management Plan (EMP). The EMP specifies concrete mitigation measures for construction and operational phases, budgets, and defines institutional responsibilities for ongoing monitoring.

v. **Recommendation Letter from the Local Level**

- The proponent must formally apply for and obtain a Recommendation Letter (**सिफारिस पत्र**) from the concerned Local Level (Rural Municipality/Municipality).
- This letter, issued in the format of Schedule 14 of the Regulations, verifies that the local government consents to the project under the condition that the proposed environmental mitigation plan is implemented.

vi. **Report Writing and Finalization**

- The IEE report is drafted in the Nepalese language (using Unicode Font size 12 as per regulatory rules) in the format prescribed in Schedule 10. It integrates all findings, baseline data, alternative analyses, local feedback, and the EMP.

vii. **Submission, Evaluation, and Approval**

- **Submission:** The finalized IEE report, along with the approved ToR, pasting evidence (Muchulkas), local recommendation letters, and evaluation fees, is submitted to the Concerned Body.
- **Technical Inquiry:** The Concerned Body reviews the report and may form an expert committee to verify the quality and feasibility of proposed mitigation measures.
- **Approval Decision:** If the report meets the necessary standard and shows no significant unmitigable adverse impacts, the Concerned Body is mandated to officially approve the IEE report within 15 days from the date of final submission.

Implementing a 1 to 50 MW hydropower project without an approved IEE report is illegal and can result in the project being halted along with a fine of up to one million Nepalese Rupees.

8. **Application of safety rules and regulations is utmost necessary during construction of hydropower project. Discuss it.**

Answer:

Safety rules and regulations are essential during the construction of hydropower projects because such projects involve large-scale excavation, tunnelling, heavy machinery, explosives, high-voltage equipment, deep water and work at considerable heights. Failure to maintain safety can result in loss of life, damage to property, project delays and financial losses.



Some major safety measures are discussed below:

i. **Compliance with laws and regulations**

- Follow applicable occupational health and safety laws, construction standards and project-specific safety requirements.
- Prepare and implement a Health, Safety and Environment (HSE) plan.
- Clearly define safety responsibilities of the owner, contractor and workers.

ii. **Safety in excavation and slope works**

- Provide proper slope stabilization, shoring and benching for excavations.
- Regularly inspect slopes, trenches and rock faces for cracks, loose rocks and landslides.
- Provide safe access and adequate drainage.

iii. **Tunnel construction safety**

- Provide adequate ventilation, lighting and drainage.
- Install suitable rock support such as rock bolts, shotcrete, steel ribs and lining.
- Monitor rock movement, groundwater and hazardous gases.
- Establish emergency escape and communication arrangements.

iv. **Blasting and explosives safety**

- Only trained and authorized personnel should handle explosives.
- Store explosives in approved and secure magazines.
- Establish exclusion zones and warning systems before blasting.
- Maintain proper records of explosives and conduct post-blast inspections.

v. **Safety around water**

- Provide barriers and warning signs around rivers, diversion channels, cofferdams and reservoirs.
- Provide life jackets and rescue equipment for workers working near water.
- Establish procedures for sudden changes in river discharge and flood events.

vi. **Machinery and equipment safety**

- Regularly inspect cranes, excavators, drilling equipment, lifting devices and vehicles.
- Only trained operators should operate heavy equipment.
- Use proper personal protective equipment (PPE) such as helmets, safety shoes, gloves, goggles and harnesses.

vii. **Electrical safety**

- Properly earth electrical equipment and temporary installations.
- Protect workers from electric shock and accidental energization.

- Use lock-out/tag-out procedures during maintenance.
- Restrict access to high-voltage areas.

viii. **Working at height**

- Provide properly designed scaffolding, guardrails and safe platforms.
- Use safety harnesses and lifelines where necessary.
- Inspect ladders, scaffolding and lifting arrangements regularly.

ix. **Fire and emergency preparedness**

- Provide fire extinguishers, alarms and emergency exits.
- Prepare emergency response plans for fire, landslide, flood, tunnel collapse, blasting accidents and electrical accidents.
- Conduct regular emergency drills and maintain first-aid and medical facilities.

x. **Training and supervision**

- Conduct safety induction and regular toolbox meetings.
- Train workers in equipment operation, emergency response and use of PPE.
- Maintain continuous safety supervision and accident reporting systems.

The recent catastrophic flood in Rasuwa clearly demonstrates why strict safety rules and disaster preparedness are essential in hydropower construction. The sudden flooding severely affected hydropower projects and infrastructure along the Bhotekoshi–Trishuli river corridor. Reports indicated damage to operational and under-construction projects, including facilities such as Rasuwagadhi, Chilime, Upper Trishuli and other projects in the affected region.

The disaster also highlighted the serious risks faced by workers at hydropower construction sites. Rescue operations focused on workers affected or trapped at project sites and tunnels, demonstrating the importance of reliable emergency communication, evacuation procedures, rescue access, and emergency preparedness.

9. If you have to develop a small hydropower project of capacity 10 MW in a cost-effective manner in a remote area of Nepal, what are the stages of study that have to be undertaken before the construction start? Discuss it.

Answer:

For a 10 MW hydropower project in a remote area, studies should be conducted progressively so that unnecessary expenditure on detailed investigations is avoided. The main objective is to select a technically feasible, economically viable, environmentally acceptable and cost-effective project before construction begins.

i. **Reconnaissance Study**

This is the first and least-cost stage of investigation.

- Collect existing topographical maps, geological maps, hydrological records and satellite imagery.
- Identify potential river reaches and possible locations for intake, powerhouse and waterways.
- Conduct a preliminary site visit.
- Assess approximate head, discharge and power potential.
- Examine accessibility, availability of construction materials and proximity to the transmission network.
- Identify major environmental and social constraints.

For a remote Nepalese site: satellite imagery and freely available DEM data should be used initially to reduce field-survey costs.

ii. **Pre-feasibility Study**

The promising alternative identified during reconnaissance is investigated in greater detail.

It includes:

- Preliminary topographical and geological survey.
- Preliminary hydrological and sediment assessment.
- Selection of possible project layouts.
- Preliminary design of intake, headrace, penstock, powerhouse and tailrace.
- Preliminary estimation of installed capacity and annual energy.
- Preliminary construction-cost estimate.
- Preliminary environmental and social assessment.
- Preliminary economic and financial analysis.

At this stage, alternative layouts should be compared to select the most economical option.

iii. Feasibility Study

A detailed feasibility study is undertaken for the selected alternative.

a. Hydrological Study:

- Establish design discharge and dependable flow.
- Analyze flood characteristics.
- Assess sediment load.
- Determine design head and energy generation.

b. Topographical study

- Detailed survey and contour mapping.
- Longitudinal and cross-sectional surveys.
- Survey of intake, waterways, powerhouse and access roads.

c. Geological and geotechnical study

- Geological mapping.
- Boreholes, trial pits and rock sampling where necessary.
- Investigation of tunnel and powerhouse foundations.
- Assessment of landslide and slope stability conditions.

d. Environmental and social study

- Environmental impact assessment as required.
- Assessment of land acquisition, local impacts and rehabilitation requirements.
- Development of environmental mitigation measures.

e. Economic and financial study

- Detailed project cost estimation.
- Economic indicators such as ENPV, EIRR and benefit-cost ratio.
- Financial indicators such as FNPV and FIRR.

iv. Detailed Project Report (DPR) and Detailed Design

After establishing feasibility, the project is developed to a construction-ready level.

The DPR should contain:

- Final project layout.
- Detailed topographical, geological and hydrological results.
- Detailed drawings and specifications.

- Detailed quantity and cost estimates.
- Construction methodology.
- Environmental and social management plans.
- Risk assessment and mitigation measures.

v. Statutory Approval and Licensing

Before construction, the necessary government approvals, environmental clearance and hydropower-related licenses must be obtained in accordance with prevailing Nepalese laws and regulations.

This stage also involves:

- Power purchase and grid-connection arrangements.
- Land acquisition/lease arrangements.
- Forest and environmental approvals, where applicable.
- Water-use and other required permissions.
- Financing arrangements.

vi. Tendering and Construction Planning

Once the project is approved and financed:

- Prepare bidding documents.
- Finalize contract packaging.
- Prequalify contractors where required.
- Invite and evaluate bids.
- Select contractors and consultants.
- Finalize construction schedule and procurement plan.
- Establish camps, access roads, communication and other temporary facilities.

10. **What are the major constraints in development of Hydropower in Nepal? What are your suggestions to overcome such constraints?**

Answer:

Nepal has enormous hydropower potential, but development is constrained by difficult geography, seasonal river flow, inadequate transmission, high financing costs, lengthy approvals, land and forest issues, weak project management, and uncertainty about electricity markets. These constraints can be reduced through integrated planning, stable policies, faster approvals, improved transmission, better financing arrangements, and stronger environmental and social safeguards.

The major constraints in development of Hydropower in Nepal are listed below:

i. Difficult topography and geology

- Nepal's rivers flow through steep, mountainous and unstable terrain.
- Projects require long tunnels, high dams, underground powerhouses, bridges and access roads.
- Landslides, rockfalls, earthquakes and difficult geological conditions increase construction risk.
- Remote project locations make transportation of machinery, cement, steel and other materials expensive.
- Limited access roads and weak logistics delay construction and increase project costs.

ii. Seasonal variation in river flow

Nepal's rivers have highly seasonal discharge. More than 80% of annual precipitation occurs during the monsoon season, while winter and pre-monsoon flows are much lower. This creates two problems:

- a. Run-of-river projects produce surplus electricity during the wet season.
- b. Generation falls substantially during the dry season, when electricity demand remains high.

Therefore, Nepal needs a balanced combination of run-of-river, peaking run-of-river, storage and pumped-storage projects.

iii. Difficulty in land acquisition and forest clearance

Hydropower projects require land for:

- Dams and headworks.
- Powerhouses and switchyards.
- Roads, tunnels and transmission lines.
- Reservoirs, construction camps and spoil-disposal areas.

Disputes commonly arise over land valuation, compensation, resettlement and the rights of affected families. Projects also frequently pass-through forest or protected areas. Obtaining approval for tree cutting and environmental clearance can take several years.

iv. Lengthy administrative procedures

A project may require approval from several agencies for:

- Survey and generation licences.
- Environmental impact assessment.
- Forest and conservation clearance.
- Land acquisition.
- Electricity generation and grid connection.
- Foreign investment and foreign-exchange approval.
- Power purchase agreements.
- Customs and import of equipment.

Overlapping responsibilities, repeated documentation and poor coordination among federal, provincial and local agencies cause delays and increase uncertainty.

v. Inadequate investment and difficult financing

Hydropower projects require large initial investment and generally take many years before generating revenue. Major financial constraints include:

- Limited domestic capital.
- High interest rates and refinancing risk.
- Limited capacity of domestic banks to finance very large projects.
- Foreign-exchange risk for imported equipment and foreign loans.
- Cost overruns caused by delays.
- Difficulty achieving financial closure.
- Uncertainty regarding power purchase agreements and electricity tariffs.

The World Bank has identified Nepal's limited fiscal space and financing requirements as important issues for scaling up hydropower and other infrastructure.

Some of the suggestions to overcome the constraints are listed below:

i. Improve infrastructure

- Develop all-weather access roads and bridges to potential hydropower sites.
- Expand and strengthen the national transmission network.
- Coordinate hydropower development with transmission planning.

ii. Improve project investigation

- Carry out detailed hydrological, geological and geotechnical investigations before construction.
- Use GIS, remote sensing and modern surveying technologies for preliminary site selection.
- Establish reliable long-term hydrological and sediment monitoring stations.

iii. Develop storage and reservoir projects

- Promote a balanced combination of run-of-river, peaking run-of-river and storage projects.
- Storage projects can help reduce the seasonal mismatch between electricity generation and demand.

iv. Streamline government procedures

- Establish a single-window approval system.
- Reduce unnecessary duplication among government agencies.
- Make licensing, environmental clearance and land-acquisition procedures faster and more transparent.

v. Improve disaster resilience

- Incorporate earthquake, landslide, flood and GLOF risks into project design.
- Establish early-warning and monitoring systems.
- Regularly inspect and maintain existing hydropower structures.

11. **What do you mean by multipurpose projects? What are merits and demerits of multipurpose projects?**

Answer:

A multipurpose project is a large water-resources development project designed to serve two or more purposes simultaneously by utilizing the available water resources in an integrated manner. For example, a reservoir may be developed for hydropower generation, irrigation, flood control, drinking water supply, navigation and fisheries at the same time.

Examples: Kulekhani Hydropower Project, Bheri- Babbai Diversion multipurpose project, Sunkoshi Marin Diversion Multipurpose project etc.



Merits of Multipurpose Projects

- i. **Multiple benefits** – A single project provides several services such as power, irrigation and flood control.
- ii. **Efficient utilization of water** – Available water resources are used more effectively.
- iii. **Hydropower generation** – Storage reservoirs can provide reliable electricity, particularly during dry seasons.
- iv. **Flood control** – Reservoir storage can reduce downstream flood peaks.
- v. **Irrigation development** – Stored water can provide irrigation during dry periods.
- vi. **Water supply** – Water can be supplied for domestic and industrial purposes.
- vii. **Regional development** – Roads, electricity, employment and other infrastructure develop around the project.
- viii. **Recreation and fisheries** – Reservoirs may provide opportunities for tourism, recreation and fisheries.
- ix. **Better economic return** – Costs can be shared among several purposes, improving overall project benefits.

Demerits of Multipurpose Projects

- i. **Very high initial cost** – Large dams and associated infrastructure require huge investment.
- ii. **Complex planning and design** – Different objectives may have conflicting water requirements.
- iii. **Long construction period** – Large multipurpose projects often take many years to complete.
- iv. **Large land requirement** – Reservoirs can submerge large areas of agricultural and forest land.
- v. **Displacement of people** – Settlements and communities may need to be relocated.
- vi. **Environmental impacts** – They can affect river ecosystems, aquatic life, forests and sediment movement.
- vii. **Sedimentation** – Reservoir storage may gradually decrease due to sediment deposition.
- viii. **Geological and seismic risks** – Large reservoirs and dams require careful assessment of landslides, earthquakes and foundation conditions.
- ix. **Difficult management** – Operation must balance competing demands such as irrigation, power generation and flood control.
- x. **Risk of catastrophic failure** – Failure of a large dam can cause severe downstream damage.